

Lecture 10: Long-Run Risks

Raul Riva

FGV EPGE

March, 2026

- Mehra-Prescott leaves too little equity premia unless risk aversion is pushed hard
- Habits and disasters fix this by changing risk prices or tail risk
- Bansal and Yaron (2004) ask a different question: can *small* persistent growth news move prices a lot?
- Long-run growth risk plus time-varying macro uncertainty is their answer
- With Epstein-Zin preferences, distant bad news becomes first-order for valuation

- Motivation: why long-run risk is not “just another $AR(1)$ ”
- A short detour: the extended consumption CAPM
- Primitives: preferences, cash-flow dynamics, and state variables
- Approximate solution via the Campbell-Shiller decomposition ([Campbell and Shiller 1988](#))
- What must be solved numerically in the full model
- Quantitative targets, plus placeholders for the key figures and tables

A Useful Detour

Notation for This Detour

$$m_{t+1} \equiv \log M_{t+1} = \theta \log \delta - \frac{\theta}{\psi} \Delta c_{t+1} - (1 - \theta) r_{w,t+1}$$

$$\tilde{x}_{t+1} \equiv x_{t+1} - E_t[x_{t+1}]$$

- M_{t+1} is the stochastic discount factor
- m_{t+1} is its log
- $r_{w,t+1}$ is the log return on the wealth portfolio
- A tilde means an innovation: the unexpected part relative to time- t information
- We will apply this to $\tilde{m}_{t+1}$, $\tilde{c}_{t+1}$, and $\tilde{r}_{w,t+1}$

Why Start with the Extended Consumption CAPM?

- Recursive utility prices more than current consumption risk
- It also prices shocks that revise continuation values
- Campbell's log-linearization makes those continuation shocks observable as “news”
- This is the clean bridge from Epstein-Zin preferences to long-run risk
- Bansal and Yaron (2004) will load exactly on that extra news term

Two Important Exercises

- Assume: the SDF and the returns of traded assets are jointly lognormal and homokedastic
- Exercise 1: show that, for any asset i , we have

$$\mathbb{E}_t[r_{i,t+1} - r_{f,t}] + \frac{1}{2}\sigma_i^2 = -\text{cov}_t(m_{t+1}, r_{i,t+1}) = \frac{\theta}{\psi}\sigma_{ic} + (1 - \theta)\sigma_{iw}$$

Two Important Exercises

- Assume: the SDF and the returns of traded assets are jointly lognormal and homokedastic
- Exercise 1: show that, for any asset i , we have

$$\mathbb{E}_t[r_{i,t+1} - r_{f,t}] + \frac{1}{2}\sigma_i^2 = -\text{cov}_t(m_{t+1}, r_{i,t+1}) = \frac{\theta}{\psi}\sigma_{ic} + (1 - \theta)\sigma_{iw}$$

- Exercise 2: show that the risk-free rate $r_{f,t+1} = \log(R_{f,t+1})$ given by:

$$r_{f,t+1} = -\log(\delta) + \frac{1}{\psi}\mathbb{E}_t[\Delta c_{t+1}] - \frac{\theta}{2\psi^2}\sigma_c^2 + \frac{(1 - \theta)}{2}\sigma_w^2 = \text{constants} + \frac{1}{\psi}\mathbb{E}_t[\Delta c_{t+1}]$$

(Hint: let $i = w$ after exercise 1)

Two Important Exercises

- Assume: the SDF and the returns of traded assets are jointly lognormal and homokedastic
- Exercise 1: show that, for any asset i , we have

$$\mathbb{E}_t[r_{i,t+1} - r_{f,t}] + \frac{1}{2}\sigma_i^2 = -\text{cov}_t(m_{t+1}, r_{i,t+1}) = \frac{\theta}{\psi}\sigma_{ic} + (1 - \theta)\sigma_{iw}$$

- Exercise 2: show that the risk-free rate $r_{f,t+1} = \log(R_{f,t+1})$ given by:

$$r_{f,t+1} = -\log(\delta) + \frac{1}{\psi}\mathbb{E}_t[\Delta c_{t+1}] - \frac{\theta}{2\psi^2}\sigma_c^2 + \frac{(1 - \theta)}{2}\sigma_w^2 = \text{constants} + \frac{1}{\psi}\mathbb{E}_t[\Delta c_{t+1}]$$

(Hint: let $i = w$ after exercise 1)

- What's the intuition of ψ ? How does it affect these rates? Why?

$$\tilde{m}_{t+1} = -\frac{\theta}{\psi}\tilde{c}_{t+1} - (1 - \theta)\tilde{r}_{w,t+1}$$

- $\tilde{r}_{w,t+1}$ is the unexpected return on the wealth portfolio
- The wealth portfolio is the claim to aggregate consumption
- Compared with the baseline CCAPM, wealth-return news is also priced
- If $\theta = 1$, we are back to the standard case

$$\tilde{m}_{t+1} = -\frac{\theta}{\psi}\tilde{c}_{t+1} - (1 - \theta)\tilde{r}_{w,t+1}$$

- $\tilde{r}_{w,t+1}$ is the unexpected return on the wealth portfolio
- The wealth portfolio is the claim to aggregate consumption
- Compared with the baseline CCAPM, wealth-return news is also priced
- If $\theta = 1$, we are back to the standard case
- Next step: rewrite $\tilde{r}_{w,t+1}$ in terms of consumption news

From Log-Linearization to News Form

$$r_{w,t+1} \approx \kappa_0 + \kappa_1 z_{t+1}^w - z_t^w + \Delta c_{t+1}, \quad z_t^w \equiv \log(P_t/C_t)$$

$$\tilde{r}_{w,t+1} = (E_{t+1} - E_t) [\Delta c_{t+1} + \kappa_1 z_{t+1}^w]$$

- This is the standard Campbell-Shiller log-linearization applied to the wealth claim
- Because the wealth portfolio pays consumption, the dividend-growth term is Δc_{t+1}
- z_t^w is known at time t , so it drops out of the innovation
- Forward-iterate z_{t+1}^w and impose the no-bubble condition
- That produces the news representation on the next slide

$$\tilde{r}_{w,t+1} = (E_{t+1} - E_t) \sum_{j=0}^{\infty} \kappa_1^j \Delta c_{t+1+j} - (E_{t+1} - E_t) \sum_{j=1}^{\infty} \kappa_1^j r_{w,t+1+j}$$

- This is the Campbell-Shiller decomposition in “cash-flow news” and “discount rate news” form
- The dividend on the wealth portfolio is consumption (see Gourio’s notes)
- First term: cash-flow news
- Second term: discount-rate news

A Return Formula in Terms of Consumption News

Let $N_{c,t+1}$ be the news about future consumption growth, so that

$$N_{c,t+1} \equiv (E_{t+1} - E_t) \sum_{j=1}^{\infty} \kappa_1^j \Delta c_{t+1+j}$$

A Return Formula in Terms of Consumption News

Let $N_{c,t+1}$ be the news about future consumption growth, so that

$$N_{c,t+1} \equiv (E_{t+1} - E_t) \sum_{j=1}^{\infty} \kappa_1^j \Delta c_{t+1+j}$$

Exercise 3: show that $(\mathbb{E}_{t+1} - \mathbb{E}_t)[r_{w,t+1+j}] = (\mathbb{E}_{t+1} - \mathbb{E}_t)[\Delta c_{t+1+j}]$ using Exs. 1 and 2

A Return Formula in Terms of Consumption News

Let $N_{c,t+1}$ be the news about future consumption growth, so that

$$N_{c,t+1} \equiv (E_{t+1} - E_t) \sum_{j=1}^{\infty} \kappa_1^j \Delta c_{t+1+j}$$

Exercise 3: show that $(\mathbb{E}_{t+1} - \mathbb{E}_t)[r_{w,t+1+j}] = (\mathbb{E}_{t+1} - \mathbb{E}_t)[\Delta c_{t+1+j}]$ using Exs. 1 and 2

Using exercise 3 we conclude:

$$\tilde{r}_{w,t+1} = \tilde{c}_{t+1} + \left(1 - \frac{1}{\psi}\right) N_{c,t+1}$$

- Intuition for the role of ψ ?

Plugging Back into the SDF

Recall the form of the SDF innovations:

$$\tilde{m}_{t+1} = -\frac{\theta}{\psi}\tilde{c}_{t+1} - (1 - \theta)\tilde{r}_{w,t+1}$$

Using what we just derived, we have:

$$\tilde{m}_{t+1} = -\gamma\tilde{c}_{t+1} - \left(\gamma - \frac{1}{\psi}\right)N_{c,t+1}$$

- **Important:** γ and ψ are not the same parameter!
- You can bump up γ without changing the IES ψ , and vice versa
- What if $\gamma = 1/\psi$?

The Extended Consumption CAPM

$$\widetilde{m}_{t+1} = -\gamma \widetilde{c}_{t+1} - \left(\gamma - \frac{1}{\psi} \right) N_{c,t+1}$$

$$\mathbb{E}_t[r_{i,t+1} - r_{f,t}] + \frac{1}{2}\sigma_i^2 = \gamma\sigma_{ic} + \left(\gamma - \frac{1}{\psi} \right) \sigma_{iN}$$

- Standard CCAPM term: covariance with current consumption shocks
- New term: covariance with news about future consumption
- Here $\sigma_{iN} \equiv \text{cov}_t(r_{i,t+1}, N_{c,t+1})$
- If $\gamma > 1$ and $\psi > 1$, bad long-run news raises marginal utility
- Long-run risk models make this second covariance quantitatively important

Primitives

- Representative agent with recursive utility ([Epstein and Zin 1989, 1991](#))
- Two aggregate cash-flow objects: consumption and dividends
- The market is a claim to dividends, not a claim to aggregate consumption
- We therefore track both $R_{a,t+1}$ and $R_{m,t+1}$
- In the full model, the state vector is (x_t, σ_t^2)

$$\mathbb{E}_t \left[\delta^\theta G_{t+1}^{-\theta/\psi} R_{a,t+1}^{-(1-\theta)} R_{i,t+1} \right] = 1$$

- G_{t+1} is gross consumption growth
- ψ is the IES, γ is risk aversion
- $\theta \equiv \frac{1-\gamma}{1-1/\psi}$
- The whole mechanism needs the Epstein-Zin wedge between γ and ψ

Why Separate $R_{a,t+1}$ and $R_{m,t+1}$?

- $R_{a,t+1}$ is the return on a claim to aggregate consumption
- $R_{m,t+1}$ is the observable market return: the aggregate dividend claim
- Solve for the consumption claim first: it pins down the IMRS
- Then price the market using dividend dynamics
- This is where leverage in dividends will matter

$$x_{t+1} = \rho x_t + \varphi_e \sigma e_{t+1}$$

$$g_{t+1} = \mu + x_t + \sigma \eta_{t+1}$$

$$g_{d,t+1} = \mu_d + \phi x_t + \varphi_d \sigma u_{t+1}$$

- x_t is the small persistent component in expected growth
- $\phi > 1$ makes dividend cash flows more exposed than consumption

What Each Shock Does

- e_{t+1} moves beliefs about future growth
- η_{t+1} is short-run consumption risk
- u_{t+1} adds dividend risk not perfectly tied to consumption
- ρ close to one makes growth news long-lived
- Tiny news today can then have large valuation effects

$$\sigma_{t+1}^2 = \sigma^2 + \nu_1 (\sigma_t^2 - \sigma^2) + \sigma_w w_{t+1}$$

- Economic uncertainty is the conditional variance of consumption growth
- ν_1 controls how persistent volatility shocks are
- w_{t+1} becomes a priced shock in the full model
- This is what generates time-varying risk premia

Why This Is a Long-Run Risk Model

- The quantity shock is small
- The persistence is large
- Duration amplifies valuation responses to growth news
- If $\psi > 1$, good long-run growth news raises wealth and asset prices
- If $\psi < 1$, the same news works in the wrong direction

Approximate Solution

Recall: Campbell-Shiller in General

$$R_{t+1} = \frac{P_{t+1} + D_{t+1}}{P_t}$$

$$r_{t+1} \approx \kappa_0 + \kappa_1(p_{t+1} - d_{t+1}) - (p_t - d_t) + \Delta d_{t+1}$$

- This is the standard log-linear return decomposition ([Campbell and Shiller 1988](#))
- P_t : ex-dividend price at time t
- D_t : payout at time t , so next period's cash flow is D_{t+1}
- R_{t+1} : gross return from t to $t + 1$
- Lowercase letters are logs: $r_t = \log R_t$, $p_t = \log P_t$, $d_t = \log D_t$
- It is a first-order expansion around the mean log price-dividend ratio

$$r_{a,t+1} = \kappa_0 + \kappa_1 z_{t+1} - z_t + g_{t+1}$$

- This is the Campbell-Shiller identity applied to the consumption claim
- Set the “dividend” equal to consumption: $D_t = C_t$
- If W_t is cum-dividend wealth, the ex-dividend price is $P_t = W_t - C_t$
- Hence $z_t = \log(P_t/C_t) = \log((W_t - C_t)/C_t)$
- Here Δd_{t+1} becomes consumption growth g_{t+1}
- There is an analogous decomposition for the market using $z_{m,t}$
- This log-linearization makes the economics transparent

Guess the Price-Consumption Ratio

$$z_t = A_0 + A_1 x_t, \quad z_{m,t} = A_{0,m} + A_{1,m} x_t$$

- In Case I, one state variable is enough
- Plug the guess into the Euler equation and match coefficients
- In the full model, we add a loading on σ_t^2

Case I: What Comes Out of the Guess

$$\mathbb{E}_t[r_{a,t+1}] = \text{const} + (1 - A_1 + \kappa_1 \rho A_1) x_t$$

$$\mathbb{E}_t[r_{m,t+1}] = \text{const} + (\phi - A_{1,m} + \kappa_{1,m} \rho A_{1,m}) x_t$$

- Substitute $z_t = A_0 + A_1 x_t$ and $x_{t+1} = \rho x_t + \varphi_e \sigma e_{t+1}$ into the Campbell-Shiller identity
- In Case I, conditional variances are constant, so conditional risk premia are constant
- Therefore the only time variation in expected returns comes from the risk-free rate
- The Euler equation implies $r_{f,t} = \text{const} + \frac{1}{\psi} x_t$

$$1 - A_1 + \kappa_1 \rho A_1 = \frac{1}{\psi}, \quad \phi - A_{1,m} + \kappa_{1,m} \rho A_{1,m} = \frac{1}{\psi}$$

$$A_1 = \frac{1 - 1/\psi}{1 - \kappa_1 \rho}, \quad A_{1,m} = \frac{\phi - 1/\psi}{1 - \kappa_{1,m} \rho}$$

- Match the coefficient on x_t on both sides and solve
- The same discount-rate slope, $1/\psi$, prices both claims
- If $\psi > 1$, then $A_1 > 0$
- More persistent x_t raises both loadings
- If $\phi > 1$, dividend claims are more sensitive than consumption claims
- This is the core “small shock, big price effect” step

$$m_{t+1} = \theta \log \delta - \frac{\theta}{\psi} g_{t+1} + (\theta - 1) r_{a,t+1}$$

- Asset prices are about covariance with m_{t+1}
- Recursive utility makes the continuation return matter directly
- Long-run news enters the SDF through $r_{a,t+1}$

$$m_{t+1} - \mathbb{E}_t[m_{t+1}] = \lambda_{m,\eta}\sigma\eta_{t+1} - \lambda_{m,e}\sigma e_{t+1}$$

- $\lambda_{m,e}$ is the price of long-run growth news
- As ρ rises, growth news becomes more permanent
- More permanence means more covariance with the IMRS
- The long-run risk premium rises with persistence

$$z_t = A_0 + A_1 x_t + A_2 \sigma_t^2$$

- A_1 is unchanged
- A_2 captures how valuation reacts to economic uncertainty
- With $\gamma > 1$ and $\psi > 1$, the paper gets $A_2 < 0$
- Higher uncertainty lowers the wealth-consumption ratio

$$\mathbb{E}_t(r_{m,t+1} - r_{f,t}) = \beta_{m,e}\lambda_{m,e}\sigma_t^2 + \beta_{m,w}\lambda_{m,w}\sigma_w^2 - \frac{1}{2}\text{var}_t(r_{m,t+1})$$

- Growth-risk and volatility-risk both earn premia
- When σ_t is high, the conditional Sharpe ratio rises
- The model can produce high premia and a low, smooth risk-free rate at the same time

What the Approximation Teaches

- Long-run growth news matters because it moves future discount rates and future cash flows
- Volatility news matters because it changes risk premia and current valuations
- The sign restrictions hinge on $\psi > 1$
- Before any numerics, the Campbell-Shiller algebra already tells us what to look for

Quantitative Checks

Figure: Long-Horizon Return Predictability

Figure: Consumption Volatility and Valuation Ratios

Table: Baseline Calibration

Numerical Implementation

How to Solve It Numerically

- Treat (x_t, σ_t^2) as the state space
- Approximate policy functions or valuation ratios with a projection basis
- Choose coefficients to minimize Euler equation errors
- Simulate at the monthly decision frequency
- Time-aggregate simulated data to annual moments before comparing to the data

How to Solve It Numerically

- Treat (x_t, σ_t^2) as the state space
- Approximate policy functions or valuation ratios with a projection basis
- Choose coefficients to minimize Euler equation errors
- Simulate at the monthly decision frequency
- Time-aggregate simulated data to annual moments before comparing to the data
- In the paper, the numerical results are close to the log-linear analytics

What to Match

- Mean and persistence of consumption and dividend growth
- Equity premium and average risk-free rate
- Volatility of the market return, the risk-free rate, and the price-dividend ratio
- Long-horizon return predictability
- The negative relation between economic uncertainty and valuation ratios

Takeaways

- The paper prices *persistent* growth news, not just bad short-run shocks
- Epstein-Zin preferences are doing essential work: $\psi > 1$ is not optional here
- Stochastic volatility gives the model time-varying premia and volatility feedback
- The approximation is simple enough to teach, but the full quantitative model is numerical
- Next step in class: decide which empirical moments we want to emphasize

Questions?

Readings and References i

- Main paper ([Bansal and Yaron 2004](#))
- Campbell-Shiller log-linearization ([Campbell and Shiller 1988](#))
- Recursive utility foundations ([Epstein and Zin 1989, 1991](#))
- Useful benchmark for comparison: habits ([Campbell and Cochrane 1999](#))
- Campbell, Chapter 6

Bansal, Ravi, and Amir Yaron. 2004. "Risks for the Long Run: A Potential Resolution of Asset Pricing Puzzles." *The Journal of Finance* 59 (4): 1481–1509. <https://doi.org/10.1111/j.1540-6261.2004.00670.x>.

Campbell, John Y., and John H. Cochrane. 1999. "By Force of Habit: A Consumption-Based Explanation of Aggregate Stock Market Behavior." *Journal of Political Economy* 107 (2): 205–51. <https://doi.org/10.1086/250059>.

- Campbell, John Y., and Robert J. Shiller. 1988. "The Dividend-Price Ratio and Expectations of Future Dividends and Discount Factors." *The Review of Financial Studies* 1 (3): 195–228. <https://doi.org/10.1093/rfs/1.3.195>.
- Epstein, Larry, and Stanley Zin. 1989. "Substitution, Risk Aversion, and the Temporal Behavior of Consumption and Asset Returns: A Theoretical Framework." *Econometrica* 57 (4): 937–69. <https://EconPapers.repec.org/RePEc:ecm:emetrp:v:57:y:1989:i:4:p:937-69>.
- . 1991. "Substitution, Risk Aversion, and the Temporal Behavior of Consumption and Asset Returns: An Empirical Analysis." *Journal of Political Economy* 99 (2): 263–86. <https://doi.org/10.1086/261750>.